

Scientific Graphics Visualisation: An Introduction to PGF/TikZ

This is the second article in a three-part series on scientific graphics visualisation. It covers PGF/TikZ software, which is free and open source. It also touches upon graphics file formats, LaTeX and PSTricks.



In the first part on this series on scientific graphics visualisation, we discussed Matplotlib, a plotting utility for Python. In this article, we will discuss PGF/TikZ, a pair of languages used for creating vector graphics. Maybe it is time to discuss, in detail, the differences between vector graphics and raster graphics.

Raster graphic images are made of pixels, which are the smallest element in a display device. If you zoom your raster image indefinitely, then at some point you will start seeing a lot of tiny dots. The quality of the image goes down as you keep zooming further. Some of the common raster image formats include JPEG, TIFF, GIF, BMP, PNG, etc. Vector images, on the other hand, are drawn with the help of vector software, with lines and shapes. The vector graphic image will always look the same, even when zoomed to very large sizes. Some of the popular vector graphics formats include AI (Adobe Illustrator), EPS (Encapsulated PostScript), SVG (Scalable Vector Graphics), etc. Due to the high quality of the images, it is always better to use vector graphics over raster graphics for technical diagrams.

A brief introduction to LaTeX

Though PGF/TikZ can be used with ConTeXt and the original TeX, it is most extensively used with LaTeX. So,

a brief introduction to LaTeX is absolutely essential before proceeding with PGF/TikZ. Due to this close relationship between the two, PGF/TikZ can be considered as a LaTeX package for programming graphics. LaTeX is a very popular cross-platform document-preparation system. It is free and open source software licensed under the LaTeX Project Public License (LPPL). LaTeX was developed by Leslie Lamport. It is based on TeX, a typesetting system developed by Donald Knuth.

Very often people consider LaTeX and TeX as one and the same, but this is not true. The fundamental difference between the two can be understood with the following example. Consider TeX as a programming language with no built-in functions and LaTeX as an extension to this language containing a lot of built-in functions. Generally, we can say that TeX is far more difficult to learn and use when compared with LaTeX. A TeX distribution such as TeX Live is often used in the Linux environment to obtain the services of LaTeX. While preparing documents with LaTeX, commands are used instead of formatting the text to get special effects like bold, italics, etc. For example, in word processing packages like LibreOffice Writer or Microsoft Word, you will see text in bold or italics; whereas in a LaTeX file you will only see commands like `\textbf` or `\textit` to provide the same